



Sandeep Malik

Principal / Staff AI Engineer

CONTACT

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Location

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CORE SKILLS

AGENTIC AI

Harness & runtime engineering, Agent & MCP design, BYO-agent/MCP platforms, orchestration, Multi-Agent, ReAct, Plan-and-Execute, Human-in-the-Loop

GENAI & RAG

LLMs, RAG (multi-query, step-back, reranking, hybrid search, LLM-as-judge, evals), Prompt Engineering, Semantic Search, SBERT

PLATFORMS

Azure OpenAI, AWS Bedrock (Guardrails), LangChain, LangGraph, LlamaIndex, Google ADK, Langflow

DATA & CLOUD

Snowflake, Databricks, Spark, Kafka, Airflow, AWS, Azure, Kubernetes, Terraform

DATABASES

Pinecone, Elasticsearch/OpenSearch, PostgreSQL, MongoDB, Delta Lake

LANGUAGES

Python, SQL, Java, Scala, Node.js

CERTIFICATIONS

- Snowflake SnowPro Core
- AWS Solutions Architect – Associate
- AWS Cloud Practitioner
- Databricks Spark 3.0 – Associate
- CCA-175 (Spark & Hadoop)
- Microsoft AZ-900

EDUCATION

M.Sc. Computer Science

Manav Bharti University, 2013

SUMMARY

AI Engineer & Data Architect with **13+ years** across 11 companies and 2 continents, now leading **agentic-AI platform & harness engineering**. A hands-on builder who ships **production-grade agentic systems** end-to-end — agents, MCP tools, RAG, and the discovery/orchestration layers that govern them — on a decade of data-platform and big-data architecture (handled up to **~2 PB** and **~200K events/sec**). Currently Principal Software Engineer & Technical Lead, Trust AI Engineering at Autodesk.

EXPERIENCE

Principal Software Engineer / Technical Lead · Autodesk

Jul 2025 – Present | Montreal, Canada

- Technical lead of a central team building an **agentic platform** (BYO-agent / BYO-MCP) — the agent & MCP catalog, discovery & orchestration layers, and the **harness/standards** for building workers.
- Shipped **3 production agents** (1 proactive, 2 reactive), **5 MCPs**, and **2 IDE plugins** used across Autodesk.
- Shift security **left** in the SDLC: MCP tools surface secure-coding & cloud standards inside AI IDEs (Cursor, Claude Code) so developers remediate in-cycle.
- Built proactive remediation agents (GitHub App) that detect repo vulnerabilities, apply fixes, and raise PRs; production observability via Opik, Splunk, CloudWatch, PagerDuty.

Data Architect / Staff Engineer · Databook Labs (USA)

Dec 2022 – Jul 2025 | Montreal, Canada · Remote

- Built the data platform **from scratch** and its **first semantic layer (early 2023)** powering Databook AI, hardening data freshness, consistency & quality.
- Spearheaded a **GPT product for strategic sales** with advanced RAG (multi-query, step-back, reranking, LLM-as-judge) to lift engagement and personalization.
- Designed **agentic AI** workflows with LangGraph, Google ADK, Conduktor & Langflow; semantic index over Pinecone & Elasticsearch on Snowflake/Databricks; led a team of data engineers.

Big Data Architect / Data Engineer · Sitecore

Mar 2022 – Dec 2022 | Montreal, Canada

- Led the core **Search** team handling **~200K events/sec**, enhancing AI-backed search and personalization for e-commerce.
- Architected a Data Lake & Warehouse for a global retailer, reducing storage costs; built Snowflake marts and automated monitoring. Stack: Elasticsearch, Spark, Kafka, Snowflake, AWS, Airflow, Docker.

Solution Architect · Cloudera

May 2021 – Mar 2022 | Montreal, Canada

- Professional Services expert onboarding customers to Cloudera Data Platform (public/private/hybrid); designed on-prem → AWS migrations and reduced cloud cost.

Software Development Specialist · Intact Insurance

Apr 2020 – Apr 2021 | Montreal, Canada

- Led a team of data engineers; built on-prem → AWS historical-data migration, refactored pipelines onto Databricks, authored Snowflake/Databricks/AWS best-practice guides.

Earlier: Data Engineer, Ericsson (2019–20 — 5G optimization on Spark/Airflow/Kubeflow/K8s, Azure) · Data Engineer, CGI (2018–19 — real-time prediction & rule engine for trade settlement) · Data Engineer, Synechron (2018, Pune) · Big Data Developer, Interglobe (2016–18, Dubai — batch + real-time pipelines over ~2 PB) · Software Engineer, MotherSumi/MIND (2014–16) · Software Developer, NIIT (2012–14).